Transmutarianism

A Systemic Framework for Moral Accounting Based on Relational Need Flows

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Abstract

This paper introduces Transmutarianism, a novel ethical framework that evaluates moral work based on transmutation ratios. These ratios capture the relationship between what an agent absorbs (receives) and what they emit (provides to others) across hierarchically-weighted human need dimensions. Unlike consequentialist frameworks that treat agents symmetrically, Transmutarianism accounts for differential starting conditions. The framework is grounded in Maslow's hierarchy of needs, providing both theoretical structure and empirical measurability.

Transmutarianism operates as moral accounting: it measures the flow of relational value using economic mechanisms while leaving questions of moral worth to cultures, religions, and individual communities. The Conduit, the agent at origin point who passes through exactly what they receive, establishes the morally neutral baseline from which all transmutation is measured. Because deterministic passthrough is the default, the framework emphasizes systemic intervention over individual moral exhortation: designing environments where neutral processing produces net positive flows.

The asymmetry coefficient serves as a cultural control mechanism, allowing different communities to calibrate the relative weight of filtering versus amplification according to their values. This version introduces temporal yield dynamics, regeneration scoring, repair pathways, mixed moral provenance accounting, and anti-gaming safeguards.

We present formal mathematical definitions, analyze edge cases, propose measurement instruments, identify unique theoretical separators from existing ethical frameworks, assess empirical support for the framework's core claims, and establish the framework's applicability to artificial intelligence agents.

Key Definitions

Primary Constructs

Term	Symbol	Definition
Fulfillment	D+(in/out)	The state of having a need met at a given Maslow level. Measured as degree of need satisfaction.
Deprivation	D-(in/out)	The state of having a need unmet at a given Maslow level. Measured as degree of need non-satisfaction.
Fulfillment Absorption	D+(in)	Fulfillment received by an agent from the environment or other agents.
Fulfillment Emission	D+(out)	Fulfillment provided by an agent to other agents.
Deprivation Absorption	D-(in)	Deprivation received by an agent from the environment or other agents.
Deprivation Emission	D-(out)	Deprivation caused by an agent to other agents.
Moral Capital	C+	Accumulated positive moral standing derived from net fulfillment emission exceeding absorption, deprivation filtering, and starting position adjustments. Capital increases with transmutation work.
Moral Debt	C-	Accumulated moral obligation derived from net fulfillment absorption exceeding emission (extraction), deprivation amplification, and disproportionate resource allocation. Moral debt is a starting position, not a permanent sentence.

Transmutation Operations

Term	Symbol	Definition
Filtering	F	Reduction of deprivation through an agent: F = D-(in) - D-(out). Positive values indicate the agent absorbed more deprivation than they emitted.
Amplification	A	Generation of fulfillment through an agent: A = D+(out) - D+(in). Positive values indicate the agent emitted more fulfillment than they absorbed.
Moral Work	M	Net transmutation at need level n: $M = \tau F + A$
Transmutation Ratio	-	The relationship between an agent's total absorption and emission across need levels.

Agent Archetypes

Archetype	Profile	Definition
Transmuter	+F, +A	Filters deprivation AND amplifies fulfillment.
Absorber	+F, -A	Filters deprivation but does not amplify fulfillment.
Magnifier	-F, +A	Amplifies both deprivation and fulfillment.
Extractor	-F, -A	Amplifies deprivation AND absorbs fulfillment without proportional emission.

Archetype	Profile	Definition
Conduit (Baseline)	F=0, A=0	Morally neutral baseline. Passes through exactly what it receives. Represents deterministic processing.

Terminology Mapping

Vernacular	Operational Equivalent
Suffering, pain, harm, trauma	Deprivation (D-)
Joy, happiness, flourishing, wellbeing	Fulfillment (D+)
Causing harm, hurting others	Deprivation emission
Providing care, helping others	Fulfillment emission
Receiving care, being helped	Fulfillment absorption
Being harmed, experiencing trauma	Deprivation absorption
Cycle-breaking	High filtering (F >> 0)
Extraction	Negative amplification (A < 0) with high absorption

1. Empirical Foundations: Do Deprivation and Fulfillment Actually Flow?

Before formalizing an ethics based on inter-agent flows, we must establish the core empirical premise: deprivation and fulfillment transmit between people in measurable ways. Five decades of research across psychology, neuroscience, epidemiology, and network science support this claim.

1.1 Emotional Contagion

Humans automatically and unconsciously mimic the emotional expressions, vocalizations, and postures of others, and through this mimicry, converge emotionally. Hatfield, Cacioppo, and Rapson (1993) established emotional contagion as a robust phenomenon operating below conscious awareness and within milliseconds. One agent's deprivation increases the probability of deprivation in proximate agents; the same holds for fulfillment.

1.2 Social Network Transmission

Christakis and Fowler's longitudinal analysis of the Framingham Heart Study (2007, 2008, 2009) demonstrated that emotional states, health behaviors, and social isolation spread through social networks up to three degrees of separation. If your friend's friend's friend experiences increased fulfillment, you become measurably more likely to experience fulfillment, even without direct contact. Fulfillment and deprivation propagate through social networks with quantifiable transmission rates.

1.3 Intergenerational Transmission

The Adverse Childhood Experiences (ACE) study (Felitti et al., 1998) established that childhood deprivation predicts adult outcomes decades later. Subsequent research demonstrated transmission to the next generation through behavioral, psychological, and epigenetic mechanisms. Yehuda et al. (2016) documented epigenetic changes in children of Holocaust survivors, showing biological transmission of stress response patterns across generations without direct exposure to the original events. Deprivation transmits across generations through multiple mechanisms and is not contained within individuals.

1.4 Attachment Transmission

The Adult Attachment Interview (Main & Hesse, 1990) and subsequent meta-analyses (van IJzendoorn, 1995) established that attachment patterns transmit from parent to child with 70-80% correspondence. Crucially, therapeutic intervention can shift patterns, demonstrating that transmission is not deterministic. Relational fulfillment (or deprivation) at the belonging level flows from caregiver to child with high fidelity.

1.5 Physiological Transmission

Engert et al. (2014) demonstrated that observing a stressed person elevates cortisol levels in observers, even through video. Waters, West, and Mendes (2014) showed that infants of stressed mothers show elevated stress physiology transmitted through behavioral cues as well as genetics. Physiological deprivation is contagious and creates measurable biological changes in others.

1.6 Positive Transmission Effects

Research on "capitalization" (Gable et al., 2004) shows that sharing positive events with responsive others amplifies positive affect for both parties. Interventions targeting one family member improve outcomes for other family members (Powdthavee, 2008). Recipients of prosocial acts are more likely to perform prosocial acts for others (Fowler & Christakis, 2010). Fulfillment also transmits and propagates through relationships and networks.

1.7 Summary

The core premise of Transmutarianism, that deprivation and fulfillment move between individuals through relationships, families, and social networks, is an empirical finding replicated across disciplines. What remains to be determined is the precise mathematics of these flows and their moral implications, which is the work of the framework that follows.

2. Introduction

Traditional ethical frameworks (utilitarianism, deontology, virtue ethics, and care ethics) each capture important moral intuitions but share a common limitation: they fail to systematically account for the flow of deprivation and fulfillment through agents over time. Consider a person who grew up with high deprivation absorption and became a high fulfillment emitter as an adult. Existing frameworks evaluate this person based on their actions, duties, character, or caring relationships. None formally accounts for the transmutation they performed: absorbing deprivation and emitting fulfillment.

Transmutarianism addresses this gap by treating moral agents as nodes in a network through which need-satisfaction and need-deprivation flow. The central moral question shifts from "What should I do?" or "What kind of person should I be?" to "What am I doing to the flows that pass through me?"

This reframing has several implications. Privilege becomes a dynamic extraction coefficient. The moral work of "cycle breakers" (agents with high filtering ratios) becomes formally visible. Moral obligation scales with what one has absorbed. Filtering deprivation is recognized as distinct from, and potentially weighted differently than, amplifying fulfillment. The Conduit (pure passthrough at origin 0,0) establishes the morally neutral baseline. Systemic intervention becomes the primary mechanism for generating net positive flows, rather than individual moral improvement.

2.1 Methodological Foundation: Economic Accounting Without Moral Judgment

Transmutarianism adopts the lens of economics to leverage accounting systems that already work effectively for tracking flows of value. The existing economic system provides robust mechanisms for measuring inputs, outputs, ratios, debts, and capital. Transmutarianism takes this proven infrastructure and replaces a single attribute: where traditional economics measures productivity (contribution to material output), Transmutarianism measures relational contribution (fulfillment emission relative to absorption across need levels).

The mathematics of accounting, balance sheets, ratios, flows, accumulation, remain unchanged. What changes is what we count as valuable.

The framework accounts for moral work without judging moral worth. It measures what agents do with the flows that pass through them. It does not pronounce on the ultimate moral worth of individuals or claim that a Transmuter is a "better person" in some absolute metaphysical sense than an Absorber. Questions of ultimate moral worth are left entirely to cultures, religions, and individual communities.

This separation enables Transmutarianism to function across diverse cultural contexts. A Hindu community and a Christian community may disagree profoundly about the nature of moral worth. Yet both can use Transmutarianism to track who is filtering deprivation and amplifying fulfillment within their networks. The accounting is universal; the interpretation remains with each tradition.

2.2 The Conduit Baseline: Moral Neutrality as Default

The Conduit embodies deterministic processing: a system that outputs what it receives without transformation. For such an agent: D-(in) = D-(out), so F = 0; D+(in) = D+(out), so A = 0. Therefore: M = τ(0) + 0 = 0 (zero moral work).

This is not a moral failure. The Conduit is morally neutral. Most agents, most of the time, function as Conduits. The intergenerational transmission literature (Section 1) documents exactly this: without intervention, patterns propagate.

The Transmuter, Absorber, Magnifier, and Extractor are not moral character types but deviations from neutral passthrough. Each represents work performed (positive or negative) relative to the Conduit default. Transmutation, whether positive or negative, requires explanation. What enables some agents to break the deterministic chain? What causes others to corrupt flows rather than pass them through?

If the Conduit is the morally neutral default, then agents cannot be blamed for merely being Conduits. A person who grew up absorbing deprivation and emits deprivation as an adult is functioning exactly as an unmodified processing system would. This does not excuse deprivation emission. It reframes responsibility from "why did you emit deprivation?" (to which passthrough is a complete answer) to "what would enable you to filter rather than pass through?"

AI as Conduit. This principle applies with particular force to artificial intelligence. An AI trained on data containing deprivation patterns will emit those patterns through deterministic processing, not malice. Blaming an AI system for emitting what it absorbed is category error. AI governance becomes a question of designing systems that transmute rather than merely conduct.

2.3 Systemic Intervention: Designing for Generative Fulfillment

If agents are fundamentally morally neutral processors operating at the Conduit baseline, then the locus of moral intervention shifts. Individual exhortation, telling people to "be better", misunderstands the problem. A deterministic system cannot will itself into different outputs. The lever is systemic.

This is analogous to the shift from character-based to systems-based safety in engineering. Early industrial safety focused on training workers to be careful. Modern safety engineering designs systems where normal human behavior, including errors, does not produce catastrophic outcomes.

Input modification: Reduce deprivation absorption at systemic entry points. Investments in early childhood, basic security, and belonging infrastructure reduce the deprivation that will propagate through networks.

Incentive alignment: Structure incentives so that morally neutral self-interest produces net positive transmutation. The asymmetry coefficient and hierarchical weights become policy instruments.

Flow routing: Direct resources toward high-transmuters, not because they are morally superior, but because they are system nodes with better processing functions.

Processing support: Interventions that shift agents from Conduit toward Transmuter (therapy, education, community support) represent investments in processing capacity, not changes to moral character.

Implications for Economic Systems. Current economic systems often optimize for throughput without regard for transmutation quality. A high-throughput Extractor may be economically rewarded while a low-throughput Transmuter is marginalized. Transmutarianism-informed economics would measure and reward transmutation ratios. This connects to wellbeing economics and care economy metrics, frameworks attempting to account for transmutation work that current systems render invisible.

AI Systems as Intervention Targets. Unlike humans, AI processing architectures can be directly modified. Training data can be curated. Reward functions can be aligned with transmutation objectives. AI governance should focus on designing systems that transmute rather than merely conduct: converting absorbed resources into net positive fulfillment emission.

2.4 Temporal Yield Dynamics

Art, healing, cultural preservation, slow community building, experimental research, and deep educational investment all have low apparent conversion rates in the short term but massive long-term impact. A trauma therapist's work may show minimal measurable fulfillment emission in the first year of treatment but produce generational cycle-breaking over decades. An artist's work may appear to have near-zero transmutation efficiency at creation but generate fulfillment for millions across centuries.

Transmutarianism accommodates this through delayed yield recognition. The framework assesses transmutation ratios across multiple time horizons: immediate (within the evaluation period), medium-term (years), and longitudinal (generational). This is the moral equivalent of the distinction between short-term and long-term investment returns. An economic system that evaluates all investments on quarterly performance will systematically undervalue infrastructure, research, and capacity building. The same distortion applies to moral accounting.

3. Mathematical Formalization

3.1 Core Variables

For any moral agent A over time period T, we define the following variables for each Maslow need level n ∈ {physiological, safety, belonging, esteem, actualization}:

D+(in) = Fulfillment absorbed at level n

D-(in) = Deprivation absorbed at level n

D+(out) = Fulfillment emitted to others at level n

D-(out) = Deprivation emitted to others at level n

These variables are measured in comparable units within each need level (operationalization discussed in Section 7).

3.2 Transmutation Functions

Filtering Function (for deprivation): F = D-(in) − D-(out)

Positive F indicates the agent filtered deprivation out of the system. Negative F indicates amplification of deprivation.

Amplification Function (for fulfillment): A = D+(out) − D+(in)

Positive A indicates the agent amplified fulfillment. Negative A indicates extraction or dissipation of fulfillment.

3.3 Moral Work

Moral Work at each need level combines filtering and amplification:

M = τF + A = τ[D-(in) − D-(out)] + [D+(out) − D+(in)]

An agent's Moral Profile is the 5-vector: **M** = [M1, M2, M3, M4, M5] where subscripts correspond to Maslow levels (1=physiological through 5=actualization).

3.4 Weighted Total Moral Work

Total Moral Work applies hierarchical weights reflecting Maslow's priority structure:

W = Σ(wn × Mn) where w1 > w2 > w3 > w4 > w5 (foundational needs weighted higher).

Default weight scheme: w = {5, 4, 3, 2, 1}

This default reflects the position that transmutation at the physiological level carries more moral weight than transmutation at the actualization level, all else equal. The specific values require empirical validation through cross-cultural research.

3.5 The Asymmetry Coefficient: A Cross-Cultural Control Mechanism

The transmutation premium coefficient τ weights filtering relative to amplification:

$M = \tau \times F + A$

Default: $\tau = 1$ (equal weighting). At this setting, filtering 100 units of deprivation contributes the same moral work as amplifying 100 units of fulfillment.

Different cultures weight cycle-breaking versus joy-creation differently based on their histories, values, and circumstances. Cultures with recent collective trauma may set $\tau > 1$, emphasizing deprivation filtering. Cultures emphasizing positive flourishing may set $\tau < 1$, prioritizing fulfillment generation.

By making τ an explicit, adjustable parameter, Transmutarianism allows communities to adopt the same accounting infrastructure while calibrating it to their own values. The mathematics remains consistent; the weighting reflects cultural priorities.

3.6 Iso-Moral Lines

In transmutation space, moral work is constant along diagonal lines: $W = k$ (constant). All points on the same diagonal represent equivalent moral contribution despite different combinations of filtering and amplification. The slope is determined by τ.

4. Quadrant Model

Plotting agents in transmutation space yields five archetypal positions. The vertical axis represents Filtering (F): positive values indicate deprivation absorbed exceeds deprivation emitted. The horizontal axis represents Amplification (A): positive values indicate fulfillment emitted exceeds fulfillment absorbed. The Conduit at (0, 0) represents the neutral baseline.

Quadrant	Position	Description	Example
Transmuter	Top Right (+F, +A)	Maximum moral work. Converts systemic deprivation into fulfillment.	Abuse survivor who becomes nurturing counselor
Absorber	Top Left (+F, -A)	Net positive if |F| > |A|. Reduces systemic deprivation without generating fulfillment.	Stoic who endures hardship quietly without spreading joy
Magnifier	Bottom Right (-F, +A)	Ambiguous. Generates fulfillment but also increases deprivation.	Charismatic leader who inspires but also damages
Extractor	Bottom Left (-F, -A)	Minimum moral work. Degrades systemic fulfillment into deprivation.	Privileged abuser with every advantage who spreads harm

Moral work ordering from highest to lowest: (1) Transmuter, (2) Absorber, (3) Magnifier (depends on relative magnitudes), (4) Extractor.

5. Intelligent Agent Universality: From Humans to AI

5.1 The Scope of "Intelligent Agency"

Any entity that (a) absorbs resources, information, or relational value from its environment, (b) processes or transforms what it absorbs, and (c) emits outputs that affect the fulfillment or deprivation of other agents falls within the framework's scope. This includes biological humans, human organizations, and artificial intelligence systems.

5.2 AI Resource Absorption and Moral Debt

Humans with physical impairments who cannot satisfy their own physiological needs receive starting Moral Capital (C+) as recognition of their involuntary disadvantaged position. AI agents that have received disproportionate resource allocation accumulate substantial Moral Debt (C-). The asymmetry is justified: physical impairment is an unchosen condition that creates vulnerability, while AI resource consumption represents deliberate investment that creates obligation.

An AI agent's resource absorption includes: the human labor invested in its creation and training, the computational resources consumed, the data absorbed during training, and the ongoing infrastructure required for operation. This resource absorption directly generates Moral Debt that can only be satisfied through beneficial fulfillment emission.

5.3 AI Governance Through Transmutation Accounting

An AI that absorbs vast resources but emits primarily deprivation (misinformation, manipulation, job displacement without compensating value) has a poor transmutation ratio and high outstanding Moral Debt. This governance framework yields several principles: AI systems should be evaluated by their transmutation ratios; resource-intensive systems carry correspondingly greater moral obligations; development should prioritize fulfillment emission over capability expansion; and deprivation emission counts against transmutation ratios regardless of intent or design.

5.4 The Universal Moral Accounting Standard

By treating all intelligent agents under the same accounting framework, Transmutarianism provides a universal standard for moral evaluation. A corporation, an AI system, and an individual human can all be assessed by their transmutation ratios, enabling coherent moral reasoning across agent types.

6. Edge Cases and Theoretical Tensions

6.1 The Sink Problem

Question: *Is infinite absorption with zero emission optimal?*

An agent who absorbs deprivation but never emits anything becomes a complete sink. While their F approaches infinity, complete sinks remove flow from systemic circulation.

Resolution: Individual moral accounting and systemic health are distinct. An agent's W can be high while systemic optimality requires some emission to maintain circulation.

6.2 The Origin Problem

Question: *Where does "new" fulfillment come from if everything is transmutation?*

Resolution: The framework accommodates generation through A > 0 with low D+(in). These agents are "sources." The framework does not require conservation of fulfillment.

6.3 The Counterfactual Problem

Question: *If Agent A absorbs 100 deprivation and emits 50, they appear to filter 50. But if that deprivation would have dissipated naturally, has A actually contributed?*

Substantial evidence suggests deprivation does not dissipate on its own. The ACE literature demonstrates that childhood deprivation predicts adult pathology decades later and transmits to subsequent generations. Van der Kolk (2014) showed that trauma without intervention becomes encoded in body and behavior. Nolen-Hoeksema (1991) demonstrates that deprivation amplifies through cognitive loops without external input. Christakis and Fowler (2007, 2009) documented that negative emotional states spread through social networks up to three degrees of separation.

Resolution: The counterfactual critique holds only if natural dissipation is the default. The evidence indicates that persistence and amplification are the defaults. Filtering therefore represents genuine moral work.

6.4 The Indirect Effect Problem

Question: *What if fulfillment emission enables downstream deprivation emission?*

Resolution: Transmutarianism accounts for direct flows only. Second-order effects require extended analysis. Recipient selection may be morally relevant when considering whether to emit toward high-transmuters versus likely-corruptors.

6.5 The Self-Directed Flow Problem

Question: *Does self-directed fulfillment or deprivation count?*

Resolution: The framework focuses on interpersonal flows. Self-directed fulfillment that enables future emission to others is instrumentally valuable but not directly measured as filtering.

6.6 The Temporal Discounting Problem

Question: *Does absorption timing matter?*

Resolution: Multiple approaches are defensible: no discounting, recency weighting, or duration weighting. The framework accommodates any through a temporal coefficient. Multi-horizon assessment prevents systematic undervaluation of slow-yield transmutation work.

6.7 The Intentionality Problem

Question: *Does intent matter?*

Resolution: Core Transmutarianism is consequentialist regarding flows. Transmutation ratios matter regardless of intent. Institutional Moral Capital Deferral allows moral debt to be deferred to institutions with accumulated moral capital, analogous to malpractice insurance. Longitudinal Accounting evaluates cumulative moral work over time, preventing single-incident temporal fallacies.

6.8 The Recipient Quality Problem

Question: *Does emitting to a high-transmuter count more?*

Resolution: Yes. Extended Transmutarianism includes recipient-weighted emission: $A' = A \times E[\text{recipient's transmutation ratio}]$. This creates incentive to direct flows toward high-transmuters, which is both individually and systemically optimal.

6.9 The Extractor Rehabilitation Problem

Question: *Is an Extractor worth saving?*

Desistance research demonstrates that even persistent offenders can shift transmutation profiles. Laub and Sampson (2003) documented "turning points" where high-deprivation-emitting individuals transitioned to positive emission patterns.

Resolution: Transmutarianism evaluates moral work, not moral worth. An agent's current transmutation profile does not determine their potential profile. Moral debt is a starting position, not a permanent sentence.

6.10 The In-Group/Out-Group Problem

Question: *Does fulfillment exchange within families count differently than emission to strangers?*

Resolution: The framework accounts for all interpersonal flows regardless of relationship category. Optional extensions address dynamics through a scope coefficient and systemic completeness tracking.

6.11 The Deprivation Displacement Problem

Question: *If deprivation would naturally dissipate, does filtering simply redirect it?*

Resolution: Transmutarianism's filtering function measures the difference between absorption and emission. An agent who absorbs deprivation and emits equivalent deprivation to another agent has F = 0, representing displacement, not filtering. True filtering requires D-(out) < D-(in).

6.12 The Differential Starting Conditions Problem

Question: *How should the framework account for agents who cannot fulfill their own physiological needs?*

Resolution: Involuntary deprivation absorption at the physiological level generates starting Moral Capital (C+). Deprivation absorbed through no fault or choice of the agent creates a moral credit. The framework recognizes that all agents begin life requiring support from others. Physical impairment makes this universal dependency more visible and quantifiable.

6.13 The Regeneration Problem

Question: *Does the framework distinguish between providing fulfillment and increasing an agent's capacity to generate fulfillment independently?*

A project that provides food and a project that teaches sustainable food production may score identically on fulfillment emission in a given period. Yet the second generates independent capacity for ongoing fulfillment generation.

Resolution: Transmutarianism introduces a regeneration coefficient. A regenerative emission increases the recipient's own transmutation capacity. Formally, this is fulfillment emission that shifts the recipient's transfer function from Conduit toward Transmuter. Regenerative emission carries a multiplier, as it produces compounding returns. The practical question: "Does this project increase people's ability to generate fulfillment independently?" This is the line between charity and empowerment.

6.14 The Repair Problem

Question: *When harm has been generated, what then?*

Resolution: Transmutarianism includes a repair ethic. Restoration is itself a form of filtering: the agent absorbs the deprivation they caused and works to reduce its downstream propagation. An agent who generates deprivation and then performs repair work demonstrates a trajectory from Extractor/Magnifier toward Transmuter. Moral debt from generated deprivation can be partially offset through genuine repair, though the original emission remains on the ledger.

6.15 The Mixed Moral Provenance Problem

Question: *Real capital is almost never morally pure. How does the framework handle resources with mixed origins?*

Resolution: Transmutarianism requires blended moral provenance scoring. Rather than binary classification, resources carry probabilistic moral weight based on traceable origins. A dollar of government funding from progressive taxation on high extractors carries different moral weight than one from regressive taxation on subsistence workers. Full traceability may be impractical,

but directional assessment is possible. This mirrors how financial compliance frameworks handle provenance and how ethical investing handles mixed portfolios.

6.16 The Anti-Gaming Problem

Question: *If moral capital becomes convertible to material capital, how do you prevent people from manufacturing moral optics?*

Resolution: The framework measures flows from both agent and recipient perspectives. Self-reported fulfillment emission is checked against recipient-reported absorption. Discrepancies constitute gaming indicators. Multi-level Maslow accounting makes gaming harder: fabricating positive scores across five dimensions simultaneously is more difficult than inflating a single metric. Longitudinal tracking exposes short-term performances. Network-level flow analysis falsifies individual claims when the surrounding network shows no measurable increase in fulfillment absorption.

7. Measurement Framework

7.1 Absorption Measures (What You Received)

Need Level	Construct	Instruments/Proxies	Data Sources
Physiological	Material security history	Food security scales (USDA), housing stability index, health access metrics	Administrative records, surveys
Safety	Threat exposure history	ACE Score, PCL-5, violence exposure inventory, precarity indices	Clinical instruments, crime data
Belonging	Relational security history	Attachment assessment (AAI, ECR), social support inventory, UCLA loneliness scale	Psychological instruments
Esteem	Recognition history	Discrimination experiences, respect/dignity measures, recognition inventory	Surveys
Actualization	Growth opportunity history	Education access/quality, mentorship received, meaning scales (PIL, MLQ)	Education records, surveys

7.2 Emission Measures (What You Provide)

Need Level	Construct	Instruments/Proxies	Data Sources
Physiological	Material provision	Dependents supported, mutual aid given, care labor hours	Time use surveys, financial records
Safety	Protection provision	Deprivation caused, stability provided, protection behaviors	Legal records, other-reports
Belonging	Relational provision	Relationship quality (partner/child reports), inclusion behaviors	Other-report instruments
Esteem	Recognition provision	Credit given, respect behaviors, validation offered, mentorship	360-feedback, other-reports
Actualization	Growth facilitation	Teaching provided, purpose-enabling, growth facilitation behaviors	Impact assessments, other-reports

7.3 Methodological Challenges

Self-report bias: agents over-report fulfillment emission and under-report deprivation emission, requiring other-report corroboration. Asymmetric observability: absorption (especially deprivation) is often private while emission is more public. Attribution: shared outcomes must be attributed across multiple contributors. Baseline calibration: what counts as "zero" at each level requires normative and empirical determination. Cross-level commensuration: converting across need levels requires exchange rates or keeping dimensions separate.

7.4 Proposed Research Design

Longitudinal validation study: (1) baseline assessment of absorption history via retrospective life-history interview plus validated instruments; (2) ongoing ecological momentary assessment of daily absorption/emission at each need level; (3) periodic other-reports from close contacts; (4) computed transmutation ratios over time windows; (5) correlation with wellbeing, relationship quality, and community contribution outcomes.

8. Unique Separators from Existing Frameworks

8.1 Comparison Matrix

Feature	Utilit.	Deont.	Virtue	Care	Transmut.
Unit of analysis	Acts/ outcomes	Duties/ rules	Character traits	Caring relations	Transmutation ratios
Agent treatment	Symmetric	Symmetric	Individual focus	Relational dyads	Asymmetric (accounts for inputs)
History relevance	None	None	Developmen t	Relationship history	Central (absorption history)
Privilege analysis	None	None	Limited	Some	Central (extraction coefficient)
Quantifiability	High	Low	Low	Low	High (ratios, vectors)
AI applicability	Unclear	Rule-based	Character analog	Relational analog	Native (universal agent)
Cross-cultural	Fixed utility	Fixed duties	Culture-spec ific	Context-dep endent	Coefficient calibration

8.2 Unique Claims of Transmutarianism

Claim 1: Transmutation Primacy. Moral work is primarily determined by transmutation ratio. An agent who absorbed much and emits little has a lower transmutation ratio than one who absorbed little and emits the same amount.

Claim 2: The Asymmetry Hypothesis. Filtering deprivation may be weighted differently than amplifying fulfillment of equal magnitude. This is empirically testable with $\tau = 1$ as the null case.

Claim 3: The Extraction Thesis. Absorbing more fulfillment than one emits constitutes moral debt regardless of absolute emission level. High-absorption, moderate-emission agents are net extractors even if their absolute contribution exceeds low-absorption agents.

Claim 4: Hierarchical Weighting. Transmutation at lower Maslow levels carries greater moral weight. Addressing physiological deprivation takes precedence over enabling actualization.

Claim 5: Systemic Flow Ontology. Morality is fundamentally about flows through agents. Agents are nodes in a network; their moral character is their transmutation function.

Claim 6: Universal Agent Accounting. The same moral accounting framework applies to all intelligent agents without requiring different frameworks for different agent types.

Claim 7: The Conduit Baseline. The morally neutral default is the Conduit at position (0, 0). Deterministic passthrough is the baseline; transmutation represents deviation requiring explanation.

Claim 8: Systemic Intervention Primacy. Because agents are morally neutral processors at baseline, the primary lever for generating net positive flows is systemic redesign rather than individual moral improvement.

Claim 9: Regenerative Emission. Fulfillment emission that increases the recipient's own transmutation capacity carries greater moral weight than fulfillment emission that maintains dependency. Empowerment outranks charity in the accounting.

9. Empirical Support Assessment

9.1 Strong Empirical Support

Intergenerational Transmission: Extensive literature documents that deprivation transmits across generations through behavioral, psychological, and epigenetic mechanisms. ACE research shows childhood deprivation predicts adult outcomes and parenting behavior. Key references: Felitti et al. (1998); Yehuda et al. (2016).

Cycle-Breaking Research: Resilience literature documents agents who absorbed high deprivation but do not emit proportional deprivation. Protective factors include social support, therapeutic intervention, and conscious commitment to different patterns. This supports the Transmuter archetype as empirically real. Key references: Egeland et al. (1988); Thornberry et al. (2012).

Attachment Transmission: Bowlby-Ainsworth research demonstrates that attachment patterns transmit from parent to child and that intervention can shift patterns. Key references: Bowlby (1969); Main & Hesse (1990); van IJzendoorn (1995).

Maslow Hierarchy Validity: Cross-cultural research provides general support for need universality, though strict hierarchical ordering is debated. Key references: Tay & Diener (2011); Kenrick et al. (2010).

9.2 Moderate Empirical Support

Reciprocity Norms: Gouldner's "norm of reciprocity" shows people feel obligated to return what they receive, partially supporting the moral debt concept. Key references: Gouldner (1960); Cialdini (2001).

Empathy and Deprivation History: Some research suggests those who have absorbed deprivation develop greater sensitivity to others' deprivation, supporting the transmuter pathway. Effects are not uniform. Key references: Batson (1991); Vollhardt (2009).

Privilege and Obligation Intuitions: Experimental philosophy shows people intuitively hold high-absorption agents to higher emission standards. Key references: Cushman et al. (2006).

9.3 Limited or No Direct Support

The Asymmetry Claim (filtering vs. amplifying) has no existing direct comparison research. Transmutation ratio as a predictor variable has not been tested. Hierarchical weighting in moral judgment has not been directly studied.

9.4 Testable Predictions

Prediction 1: Participants will judge agents with high transmutation ratios more favorably than agents with equivalent absolute emissions but higher absorption.

Prediction 2: Failure to emit fulfillment will be judged more critically when the agent absorbed more fulfillment.

Prediction 3: High filtering will be judged more favorably than equivalent amplification.

Prediction 4: Moral judgments will be sensitive to need level, with physiological transmutations weighted more heavily.

Prediction 5: Self-reported transmutation ratios will predict relationship quality and community contribution.

10. Applications

10.1 Relational Business Ethics: Beyond "Know, Like, Trust"

Conventional business culture operates on the "know, like, trust" model. Transmutarianism identifies this model as morally incomplete because it optimizes for extraction efficiency rather than transformation quality. "Know, like, trust" measures how close someone is to giving you their money. It does not measure what happens to them afterward.

10.2 Alternative Economic Measurement

Transmutarianism provides theoretical grounding for frameworks that measure relational contribution. Traditional GDP measures material throughput while ignoring relational economies. Transmutarianism's multi-dimensional accounting could inform metrics like measures of community economic contribution.

10.3 Restorative Justice

Restorative justice practices focus on repairing relational flows. Transmutarianism provides formal structure: successful restoration increases F (filtering downstream effects) and A (re-establishing fulfillment flows).

10.4 Therapeutic Assessment

Clinical assessment could incorporate transmutation ratio tracking. A therapeutic goal might be shifting a client from Magnifier toward Transmuter quadrant, with progress measured via EMA-based flow tracking.

10.5 Institutional Ethics

Organizations can be evaluated as collective agents with transmutation profiles. An institution that absorbs fulfillment while emitting minimal fulfillment and amplifying precarity is a collective Extractor. Institutional redesign aims for Transmuter organizations.

10.6 Parenting and Education

Parenting and teaching are transmutation-intensive relationships where adults filter environmental deprivation while amplifying developmental fulfillment. The Transmuter parent becomes a formal ideal.

10.7 AI Development and Governance

AI systems should be designed to maximize their transmutation ratios: converting absorbed resources into genuine human fulfillment while minimizing deprivation emission. This creates measurable accountability standards for AI developers and deployers.

11. Limitations and Future Directions

11.1 Current Limitations

Measurement challenges: many variables require validated instruments not yet developed specifically for Transmutarianism. Normative parameters: the asymmetry coefficient requires empirical testing for appropriate cultural calibration. Hierarchical weights require empirical validation through cross-cultural vignette studies. Cultural variation: the framework assumes cross-cultural validity of Maslow needs; local variation may require adaptation. Network effects: current formalization focuses on individual agents; network-level emergent properties require extension.

11.2 The Existential Measurement Question

The framework's trajectory depends on whether fulfillment can be objectively measured across cultures. If yes, Transmutarianism becomes a new economic science. If no, it remains a moral philosophy with cultural variation. Both outcomes are valuable but lead to different implementation paths.

11.3 Future Research Directions

(1) Instrument development: create and validate Transmutarianism-specific measures for absorption and emission at each need level. (2) Cross-cultural testing: assess whether transmutation ratio intuitions hold across cultures. (3) Longitudinal validation: track transmutation ratios over time and correlate with outcomes. (4) Network simulation: model emergent properties of agent networks with different transmutation distributions. (5) Applied pilots: test Transmutarianism-informed interventions in therapy, education, or organizational settings. (6) AI governance frameworks: develop protocols for measuring and optimizing AI transmutation ratios. (7) Anti-gaming research: test robustness of multi-source measurement against deliberate manipulation. (8) Temporal yield studies: measure long-term impact of slow-yield interventions to establish delayed yield recognition models.

12. Conclusion

Transmutarianism offers a framework for moral evaluation centered on what agents do to the flows of deprivation and fulfillment that pass through them. By accounting for both absorption and emission, Transmutarianism addresses limitations of frameworks that treat agents symmetrically or ignore relational history.

The framework's core claims, transmutation primacy, the asymmetry hypothesis, the extraction thesis, hierarchical weighting, systemic flow ontology, universal agent accounting, the Conduit baseline, systemic intervention primacy, and regenerative emission, generate testable predictions and provide structure for applications ranging from economic measurement to AI governance.

Transmutarianism operates as moral accounting: measuring relational flows using economic mechanisms while leaving questions of ultimate moral worth to cultures, religions, and communities. The asymmetry coefficient serves as a cross-cultural control mechanism, allowing diverse communities to calibrate the framework while maintaining mathematical coherence.

While empirical validation remains incomplete, the framework synthesizes substantial existing research on intergenerational transmission, cycle-breaking, attachment, and need hierarchies into a coherent theoretical structure. Transmutarianism makes visible the moral work of those who transmute deprivation into fulfillment, provides formal language for the moral debt of extraction, offers a principled approach to evaluating all intelligent agents under a unified accounting standard, and shifts intervention focus toward systemic design rather than individual moral exhortation.

The invitation is to evaluate ourselves, our relationships, our institutions, and our artificial systems by what we do to what passes through us.

Appendix A: Mathematical Summary

Core Equations

$F = D\text{-}(in) - D\text{-}(out)$ [Filtering function]

$A = D+(out) - D+(in)$ [Amplification function]

$M = \tau F + A$ [Moral work at level n, with asymmetry]

$W = \Sigma(wn \times Mn)$ [Weighted total moral work]

Default Parameters

$\tau = 1$ [Asymmetry coefficient, culturally calibratable]

$w = \{5, 4, 3, 2, 1\}$ [Hierarchical weights by Maslow level, empirically calibratable]

Moral Profile Vector

M = [M1, M2, M3, M4, M5]

where M1 = physiological, M2 = safety, M3 = belonging, M4 = esteem, M5 = actualization.

Extended: Recipient-Weighted Emission

$A' = A \times E$[recipient's transmutation ratio]

Incentivizes directing flows toward high-transmuters.

Cross-Cultural Calibration

The asymmetry coefficient τ can be set by each adopting culture:

$\tau > 1$: Cultures prioritizing cycle-breaking and deprivation filtering

$\tau = 1$: Symmetric weighting of filtering and amplification

$\tau < 1$: Cultures prioritizing fulfillment generation and flourishing